MASTER OF SCIENCE IN ENGINEERING

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Advanced Statistical Data Analysis
Worksheet Week 1

Exercise 1: A soft drink bottler is analysing the vending machine service routes in his distribution system. He is interested in predicting the amount of time required by the route driver to service the vending machines in an outlet. This service activity includes stocking the machine with beverage products and minor maintenance or housekeeping. The industrial engineer responsible for the study has suggested that the two most important variables affecting the delivery time (`Time`) are the number of cases of product stocked (`volume`), and the distance walked by the route driver (`distance`). The engineer has collected 25 observations on delivery time which are given in the datset `Softdrink.dat`.

a) Fit the regression model

$$\text{Time}_i = \beta_0 + \beta_1 \cdot \text{volume}_i + E_i, \qquad i = 1, \ldots, n = 25,$$

where the errors E_i are independent, Gaussian distributed with expectation zero and variance σ^2, to the data by the least squares method.

Assess the quality of this fit by a residual and sensitivity analysis (including the simulation approach).

b) Apply Tukey's first-aid transformations to remedy model inadequacies.

c) Write down the model fitted in **??** in the original scale: $\text{Time}_i = \ldots$.

d) Extend the regression model of **??** suitably by the second explanatory variable `distance`.

Exercise 2: A research engineer is investigating the use of a windmill to generate electricity. He has collected data on the DC output (`DC.output`) from his windmill and the corresponding wind velocity ([meter per second], `velocity`) which are given in the datset `Windmill.dat`.

a) We start by fitting the regression model

$$\text{DC.output}_i = \beta_0 + \beta_1 \cdot \text{velocity}_i + E_i\,,$$

where the errors E_i are independent, Gaussian distributed with expectation zero and variance σ^2. Assess the quality of this fit by a residual and sensitivity analysis.

b) Apply Tukey's first-aid transformations. Can you remedy model inadequacies?

c) The theory of windmill operation suggests the following relationship:

$$\text{DC.output}_i = \beta_0 + \beta_1 \frac{1}{\text{velocity}_i} + E_i \quad \text{with } E_i \text{ indep.} \sim \mathcal{N}\left\langle 0, \sigma^2 \right\rangle .$$

Does this transformation help to remedy model inadequacies?

d) How do you interpret the parameters β_0 and β_1 of the regression model given in ?? in a diagram `DC.output` against `velocity`?

Exercise 3: These data relate to the construction of 32 light water reactor (LWR) plants constructed in the U.S.A in the late 1960's and early 1970's. The data was collected with the aim of predicting the cost of construction of further LWR plants. Six of the power plants had partial turnkey guarantees and it is possible that, for these plants, some manufacturers' subsidies may be hidden in the quoted capital costs. Thus, we are interested in how this effect affects the costs of construction adjusted by the other influencing effects. The dataset in `NPScosts.dat` contains the following columns:

Name	Description
`cost`	The capital cost of construction in millions of dollars adjusted to 1976 base.
`date`	The date on which the construction permit was issued. The data are measured in years since January 1 1990 to the nearest month.
`t1`	The time between application for and issue of the construction permit.
`t2`	The time between issue of operating license and construction permit.
`cap`	The net capacity of the power plant (MWe).
`pr`	A binary variable where 1 indicates the prior existence of a LWR plant at the same site.
`ne`	A binary variable where 1 indicates that the plant was constructed in the north-east region of the U.S.A.
`ct`	A binary variable where 1 indicates the use of a cooling tower in the plant.
`bw`	A binary variable where 1 indicates that the nuclear steam supply system was manufactured by Babcock-Wilcox.
`cum.n`	The cumulative number of power plants constructed by each architect-engineer.
`pt`	A binary variable where 1 indicates those plants with partial turnkey guarantees.

a) Consider Tukey's first-aid transformations where sensible (think carefully about this issue).

b) Fit a regression model including all explanatory variables. Does partial turnkey guarantee affect the costs of the plants significantly on the 5% level?

c) Assess the quality of the fitted regression model by a residual and sensitivity analysis.